Octopus: Alleviating Hallucination via Dynamic Contrastive Decoding

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Abstract

Large Vision-Language Models (LVLMs) have demonstrated remarkable capabilities in multimodal understanding and generation, yet they remain prone to object hallucination—fabricating objects, attributes, or relations not grounded in the visual input. Contrastive Decoding (CD) has emerged as a promising inference-time remedy, but existing CD methods apply a fixed strategy uniformly across all samples and generation steps, ignoring the fact that hallucination causes are fundamentally hybrid*: different samples and even different tokens within the same generation can suffer from distinct failure modes, such as over-reliance on language priors, loss of visual information, or attention bias toward irrelevant image regions. We propose* ***Octopus****, a framework that learns to dynamically select the most appropriate CD strategy at* every decoding step*. Octopus augments a frozen LVLM with a lightweight transformer block containing a learnable decision token (the "eye") that attends to the model's hidden states and outputs an action over a set of CD strategies ("tentacles"): VCD, M3ID, AVISC, or a null action. The Octopus module is trained via Direct Preference Optimization (DPO) on preference pairs constructed by sampling action sequences and ranking them with hallucination metrics, requiring no LVLM retraining or external reward models. Extensive experiments on LLaVA-1.5-7B and InstructBLIP across generative benchmarks (MMHalBench, Object HalBench, AMBER) and discriminative benchmarks (AMBER, POPE-MSCOCO) demonstrate that Octopus achieves state-of-the-art hallucination reduction—improving MMHalBench score by 64% (1.59→2.61) and reducing AMBER CHAIR by 40% (8.0→4.8) over the baseline—while remaining plug-and-play and readily extensible to new CD strategies.*

1. Introduction

Large Vision-Language Models (LVLMs) [2, 6, 15, 16, 40] have achieved remarkable progress across diverse multimodal tasks, from image captioning and visual question answering to scene understanding and dialogue. Despite their impressive performance, LVLMs are susceptible to *hallucination* [12, 22, 29]: they generate plausible yet factually incorrect descriptions that include objects, attributes, or relationships not present in the input image. This phenomenon poses significant risks for real-world deployment, where factual accuracy is critical.

Hallucination in LVLMs arises from multiple distinct causes. Language priors learned during pretraining can dominate over visual signals, causing the model to generate statistically frequent but visually ungrounded content. As generation progresses, the model's reliance on the visual prompt diminishes, leading to increasingly hallucinated outputs. Furthermore, LVLMs often exhibit *attention bias*—disproportionately attending to irrelevant background or non-object image regions ("blind tokens") rather than salient visual content. These causes are not mutually exclusive; they may co-occur within a single sample or manifest at different generation steps.

Contrastive Decoding (CD) [7, 11, 33] has emerged as an effective *training-free* approach to mitigate hallucinations at inference time. The core idea is to contrast the output distribution of a standard forward pass with that of a modified forward pass designed to emphasize a particular failure mode. VCD [11] contrasts outputs from clean and noise-distorted visual inputs to suppress language priors; M3ID [7] amplifies the mutual information between the generated token and the visual prompt to recover visual grounding; AVISC [33] identifies attention-biased "blind tokens" and performs contrastive decoding to recalibrate visual attention. Each strategy effectively addresses a *specific* hallucination cause.

However, a fundamental limitation of existing CD methods is their *one-size-fits-all* nature: a single strategy is applied uniformly to all samples and all generation steps. As illustrated in Fig. 2, each CD method is effective only on a *subset* of samples—their effective regions overlap only partially, and no single strategy dominates across the board. Moreover, as shown in Fig. 3, the hallucination cause can shift *across tokens within a single generation*, meaning a fixed strategy is inherently suboptimal.

This observation motivates our key insight: **hallucination causes are hybrid across both samples and generation steps**, and therefore an effective CD approach must dynamically adapt its strategy at each decoding step based on the current multimodal context.

To this end, we propose **Octopus**, a dynamic contrastive

decoding framework that learns to select the most appropriate CD strategy per token. Inspired by the octopus—an animal with a central brain (the "eye") directing multiple semi-independent arms ("tentacles")—Octopus augments a frozen LVLM with a lightweight transformer block containing a learnable *decision token* that attends to the model's hidden states and outputs one of four actions: apply VCD, apply M3ID, apply AVISC, or take no action (null). At each generation step, the selected tentacle constructs its contrastive input and adjusts the output logits accordingly. The LVLM weights remain entirely frozen; only the Octopus module is trained.

Training Octopus is accomplished via Direct Preference Optimization (DPO) [21]. We sample multiple action sequences for a given input and rank them using CHAIR [22] or task-appropriate hallucination metrics, constructing preference pairs without requiring a reward model or any modification to the LVLM. This makes Octopus both efficient and broadly applicable across different LVLM architectures.

Our contributions are as follows:

- We empirically demonstrate that hallucination causes are hybrid at both the sample and token levels, and that single CD strategies are effective only on subsets of cases.
- We propose Octopus, a dynamic per-token CD controller consisting of a decision token and an action space over multiple CD strategies (VCD, M3ID, AVISC, null), trained via DPO while the LVLM remains frozen.
- We introduce a DPO-based training pipeline that constructs preference pairs by sampling action sequences and ranking with hallucination metrics, requiring no reward models or LVLM retraining.
- Octopus achieves state-of-the-art hallucination reduction across generative and discriminative benchmarks on LLaVA-1.5-7B and InstructBLIP, and demonstrates strong generalizability and extensibility.

2. Related Work

Large Vision-Language Models. LVLMs integrate a visual encoder with a large language model (LLM) to enable multimodal understanding and generation [1, 2, 6, 15, 16, 40]. LLaVA [16] and LLaVA-1.5 [15] establish strong baselines using CLIP visual features projected into the LLM token space. InstructBLIP [6] employs a Q-Former architecture for visual-language alignment. Despite strong performance, all such models suffer from hallucinations [12, 22, 29], generating plausible but visually ungrounded descriptions.

Hallucination Mitigation via Fine-Tuning. Several approaches mitigate hallucinations by updating LVLM parameters. RLHF-based methods [19, 25, 36] train models with human preference feedback. DPO-based approaches [23, 28, 38, 39] construct preference pairs to directly optimize for reduced hallucination without explicit reward models. HACL [10] uses hallucination-augmented contrastive learning. EOS [37] and POVID [39] employ preference fine-tuning to align model outputs with factual visual content. Methods such as [14, 27, 35] additionally curate or augment training data to reduce hallucination. Detection-based methods [8, 34] proactively identify and suppress hallucinated content. While effective, all fine-tuning approaches modify the LVLM and thus cannot be easily transferred across different model families without retraining.

Training-Free Hallucination Mitigation. A complementary line of work targets hallucination at inference time without modifying model parameters. OPERA [9] applies an over-trust penalty and retrospection-allocation strategy to suppress repetitive or incorrect outputs. VCD [11] introduces Visual Contrastive Decoding, contrasting logits from clean and noise-distorted visual inputs to suppress language priors. M3ID [7] amplifies the mutual information between generated tokens and the visual prompt to restore visual grounding. AVISC [33] identifies "blind tokens" in image attention and performs contrastive decoding to recalibrate attention. ICD [30] contrasts standard and instruction-disturbed outputs. LCD [17] uses layer-wise contrastive decoding. DoLA [5] contrasts different transformer layers to improve factuality. ConVis [20] and HALC [4] apply focal or adaptive contrastive strategies. RITUAL [32] uses random image transformations as anti-hallucination levers. These methods each address a specific hallucination mechanism and apply uniformly across all decoding steps—our work addresses their fundamental limitation by dynamically selecting among them per token.

Preference Optimization. DPO [21] eliminates the need for a separate reward model by directly optimizing the policy using preference pairs. PPO [24] and variants [19] provide RL-based alternatives. SimPO [18] offers a reference-free reward formulation. We adopt DPO for its simplicity, stability, and effectiveness in training our decision module without LVLM retraining.

3. Preliminary

3.1. LVLM Generation and Hallucination

An LVLM generates a sequence of tokens $\mathbf{y} = (y_1, y_2, \ldots, y_T)$ conditioned on a visual input $\mathbf{v}$ and a text prompt $\mathbf{q}$. At each step t, the model computes a probability distribution over the vocabulary:

$$p(y_t \mid \mathbf{v}, \mathbf{q}, y_{<t}) = \text{softmax}(\mathbf{z}_t), \tag{1}$$

where $\mathbf{z}_t$ are the logits produced by the language model head given hidden state $\mathbf{h}_t$. Hallucination occurs when y_t is not

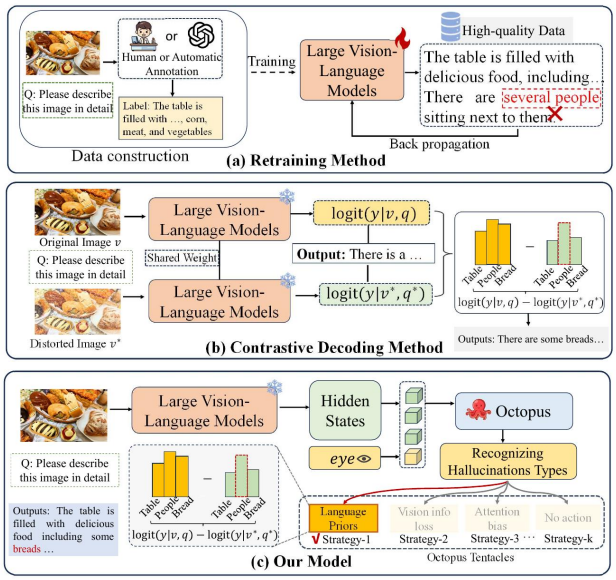

Figure 1. Paradigm comparison of different hallucination alleviation methods. (a) Fine-tuning methods update LVLM parameters. (b) Fixed CD methods apply a single strategy uniformly. (c) Octopus dynamically selects the most suitable CD strategy per token.

grounded in $\mathbf{v}$, typically due to language prior dominance, reduced visual attention, or attention bias.

3.2. Contrastive Decoding (CD)

Contrastive Decoding [11] adjusts the output logits by contrasting the standard output distribution with that of a modified forward pass designed to amplify a specific failure mode. Given a contrastive input $\tilde{\mathbf{v}}$ (or $\tilde{\mathbf{q}}$), the adjusted logits are:

$$\mathbf{z}_t^{\text{CD}} = \mathbf{z}_t - \alpha \cdot \tilde{\mathbf{z}}_t, \tag{2}$$

where $\tilde{\mathbf{z}}_t = \text{LM}(\tilde{\mathbf{v}}, \mathbf{q}, y_{<t})$ are the contrastive logits and α is the contrastive strength.

VCD [11] sets $\tilde{\mathbf{v}} = \mathbf{v} + \boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon}$ is Gaussian noise, so that $\tilde{\mathbf{z}}_t$ represents language-prior-dominated output. Subtracting it from $\mathbf{z}_t$ reduces over-reliance on statistical priors.

M3ID [7] constructs the contrastive input by marginalizing out the image ($\tilde{\mathbf{v}} = \emptyset$), effectively amplifying the mutual information between the generated text and the visual prompt, favoring tokens strongly grounded in the image over those driven by the language prior alone.

AVISC [33] identifies "blind tokens"—image tokens receiving disproportionate attention that are irrelevant to the query (e.g., background regions)—and masks them to construct $\tilde{\mathbf{v}}$, then applies CD to reduce the influence of attention-biased outputs.

3.3. Motivation: Hybrid Hallucination Causes

To understand the limitations of fixed CD strategies, we analyze their effectiveness across samples and generation steps.

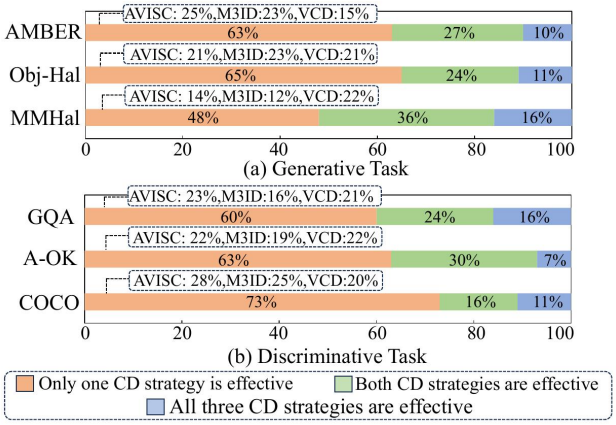

Figure 2. The proportion of effective samples using different CD methods for (a) Generative Task and (b) Discriminative Task. Each CD strategy addresses a distinct subset of samples; only a small fraction benefits from all three simultaneously.

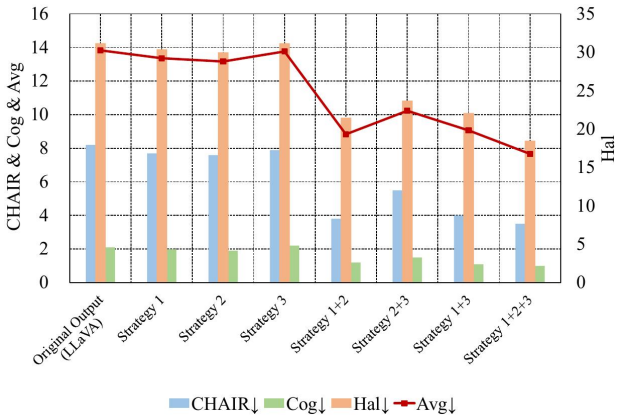

Figure 3. Token-level hallucination quantitative evaluation, illustrating that hallucination causes shift across generation steps within a single sample.

As shown in Fig. 2, for both generative and discriminative tasks, VCD, M3ID, and AVISC each improve outcomes on a different subset of samples, with only a small overlap among all three. This confirms that *no single CD strategy dominates universally*.

Furthermore, Fig. 3 reveals that even within a single generation sequence, the hallucination cause varies across tokens: a token may be driven by language priors early in generation, but by visual information loss or attention bias at later steps. This underscores the need for *token-level adaptive* CD strategy selection.

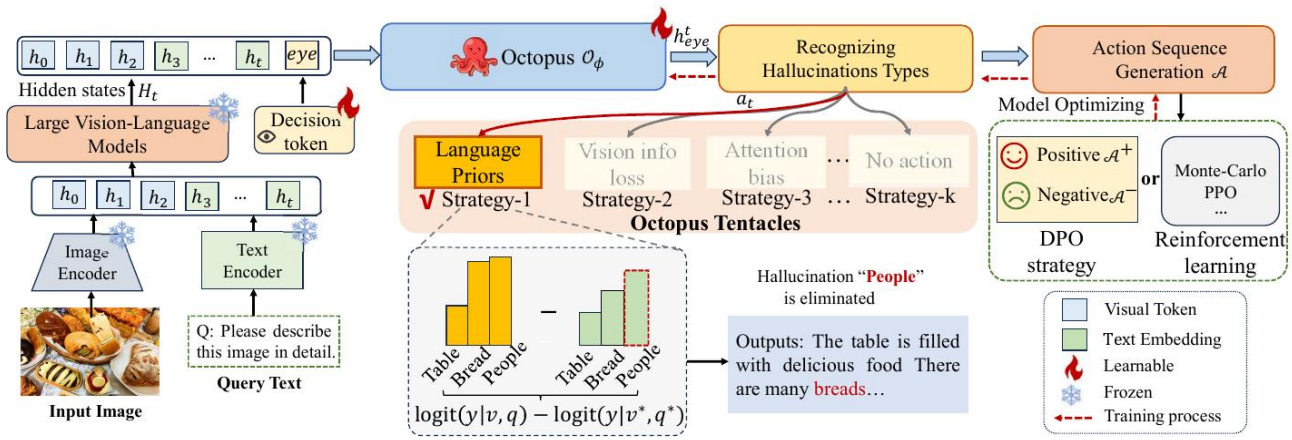

Figure 4. **Overview of Octopus.** A lightweight transformer block with a learnable decision token ("eye") attends to the frozen LVLM's hidden states and selects among CD strategies ("tentacles": VCD, M3ID, AVISC, null) at each generation step. Training uses DPO on preference pairs built from sampled action sequences ranked by hallucination metrics.

4. Our Method: Octopus

4.1. Framework Overview

As illustrated in Fig. 4, Octopus consists of two components: (1) a frozen pre-trained LVLM that produces hidden states $\mathbf{H}_t \in \mathbb{R}^{L\times d}$ at each decoding step t, where L is the sequence length and d is the hidden dimension; and (2) the Octopus module—a lightweight transformer block with a learnable decision token $\mathbf{e} \in \mathbb{R}^d$ (the "eye") that processes the hidden states and produces an action selection over the CD strategies.

The action space $\mathcal{A} = \{a_1, a_2, a_3, a_0\}$ corresponds to VCD, M3ID, AVISC, and null (no CD), respectively. At each step t, the Octopus module selects action $a_t^* \in \mathcal{A}$ and the selected tentacle adjusts the output logits via Eq. 2, from which the next token y_t is sampled.

4.2. Decision Token and Action Selection

The Octopus module appends the decision token $\mathbf{e}$ to the hidden state sequence and processes the concatenated sequence through a transformer block [26]:

$$[\mathbf{H}_t', \mathbf{e}_t'] = \text{TransformerBlock}([\mathbf{H}_t; \mathbf{e}]), \tag{3}$$

where $[\cdot;\cdot]$ denotes concatenation and $\mathbf{e}_t'$ is the updated decision token representation. A linear classification head maps $\mathbf{e}_t'$ to logits over the action space:

$$\pi_\theta(a_t \mid \mathbf{H}_t) = \text{softmax}(\mathbf{W}\mathbf{e}_t' + \mathbf{b}), \tag{4}$$

where $\mathbf{W}$ and $\mathbf{b}$ are learnable parameters of the classification head. The policy π_θ selects the action with the highest probability:

$$a_t^* = \arg\max_{a\in\mathcal{A}} \pi_\theta(a \mid \mathbf{H}_t). \tag{5}$$

This cross-attention-based design allows the decision token to dynamically attend to all hidden states, capturing global context about the current generation state across both visual and language tokens.

4.3. CD Tentacles

Once action a_t^* is selected, the corresponding tentacle constructs its contrastive input and adjusts logits. The final token probability is:

$$p^*(y_t) \propto \exp\big(\log p(y_t \mid \mathbf{v}, \mathbf{q}, y_{<t}) - \alpha \log p(y_t \mid \tilde{\mathbf{v}}, \mathbf{q}, y_{<t})\big), \tag{6}$$

where $\tilde{\mathbf{v}}$ is the tentacle-specific contrastive input. For a_0 (null), the standard logits are used without modification. The contrastive strength α can be set per tentacle or uniformly.

4.4. DPO Training

Training Octopus requires learning a policy over action sequences $\mathbf{a} = (a_1, \ldots, a_T)$ given an input $(\mathbf{v}, \mathbf{q})$. We frame this as a preference optimization problem and apply DPO [21], avoiding the need for an explicit reward model.

Preference Pair Construction. For each training sample $(\mathbf{v}, \mathbf{q})$, we sample K action sequences $\{\mathbf{a}^{(k)}\}_{k=1}^K$ from the current policy π_θ, generate the corresponding text output $\mathbf{y}^{(k)}$ using the selected CD strategies, and score each output using CHAIR [22] (for generative tasks) or accuracy-based metrics (for discriminative tasks). The action sequence achieving the best score is designated the *preferred* sequence $\mathbf{a}^+$, and the worst-scoring sequence is the *dispreferred* sequence $\mathbf{a}^-$, forming a preference pair $(\mathbf{a}^+, \mathbf{a}^-)$.

DPO Objective. Given preference pairs, DPO optimizes:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(\mathbf{a}^+ \mid \mathbf{H})}{\pi_{\text{ref}}(\mathbf{a}^+ \mid \mathbf{H})} - \beta \log \frac{\pi_\theta(\mathbf{a}^- \mid \mathbf{H})}{\pi_{\text{ref}}(\mathbf{a}^- \mid \mathbf{H})}\right)\right], \tag{7}$$

where π_{ref} is a reference policy (initialized from a uniformly random action policy), β controls the KL penalty, and $\mathbf{H}$ denotes the full sequence of hidden state trajectories. This formulation trains π_θ to prefer action sequences that produce less hallucinated outputs, without requiring LVLM parameter updates or an explicit reward model.

LVLM Frozen. Critically, the LVLM's parameters remain frozen throughout training. Only the Octopus module (transformer block, decision token, classification head) is optimized. This ensures that Octopus is *plug-and-play*: it can be attached to any compatible LVLM without altering its pretrained capabilities, and new CD tentacles can be integrated by simply extending the action space.

5. Octopus Tentacles

We briefly summarize each CD tentacle integrated into Octopus:

Strategy 1 (VCD) [11]: Constructs a distorted visual input by adding Gaussian noise $\boldsymbol{\epsilon}$ via a diffusion-based noise schedule. The logit difference suppresses statistical language priors that dominate when visual content is degraded.

Strategy 2 (M3ID) [7]: Amplifies the visual conditioning by increasing the weight of image tokens in the attention computation, maximizing the mutual information between the visual input and generated tokens to recover visual grounding that diminishes over long generation sequences.

Strategy 3 (AVISC) [33]: Identifies blind tokens—image patch tokens that receive disproportionately high attention despite being irrelevant to the query—by analyzing layer-wise attention distributions. These tokens are masked to create the contrastive visual input, reducing hallucinations caused by attention bias.

Null Action: Passes the standard logits without any contrastive modification, preserving generation quality for tokens where no hallucination correction is needed.

The modular design makes it straightforward to add new tentacles in the future, simply by extending the action space and retraining the decision module.

6. Experiment

6.1. Setup

Models. We conduct experiments primarily on **LLaVA-1.5-7B** [15] and report generalization results on **InstructBLIP** [6] (Vicuna-7B backbone) to demonstrate cross-architecture applicability.

Benchmarks. We evaluate on five benchmarks covering both generative and discriminative tasks:

- **MMHalBench** [25]: Measures hallucination rate (HalRate, ↓) and response quality score (↑) using GPT-4 evaluation.
- **Object HalBench**: Measures CHAIR$_s$ and CHAIR$_i$ metrics [22] (↓) for object hallucination in generated captions.
- **AMBER** [29] (Generative): An LLM-free multidimensional benchmark measuring CHAIR (↓), Coverage (↑), HalRate (↓), and Cognition (↓) for captioning tasks.
- **AMBER** (Discriminative): Measures accuracy and F1 for existence/attribute/relation hallucination discrimination.
- **POPE-MSCOCO** [12]: Evaluates object existence hallucination via yes/no questions under random, popular, and adversarial sampling strategies on MSCOCO [13].

Baselines. We compare against: the LVLM base model; fixed CD methods (VCD [11], M3ID [7], AVISC [33], ICD [30], OPERA [9], LCD [17], ConVis [20]); and fine-tuning methods (HACL [10], POVID [39], EOS [37], HA-DPO [38], HALVA [23]) as reference upper bounds (not directly comparable since they modify model weights).

6.2. Main Results: Generative Tasks

Table 1 reports results on generative benchmarks. On LLaVA-1.5-7B, Octopus achieves the best performance across all metrics on all three benchmarks. On MMHal-Bench, Octopus improves the score from 1.59 (base) to **2.61** (+64%) and reduces HalRate from 0.72 to **0.50**—surpassing all three individual CD strategies (VCD: 1.96, M3ID: 2.14, AVISC: 2.19). On Object HalBench, CHAIR$_s$ drops from 25.0 to **20.8** and CHAIR$_i$ from 9.2 to **6.6**, outperforming the best single method (AVISC: 22.1/7.8). On AMBER, CHAIR reduces from 8.0 to **4.8** (a 40% reduction), Coverage increases from 44.5 to **49.2**, HalRate drops from 31.0 to **23.4**, and Cognition from 2.2 to **1.2**. Each individual CD strategy falls short: VCD achieves CHAIR 6.7, M3ID 6.0, AVISC 6.3—all substantially worse than Octopus's 4.8—confirming that dynamic, per-token strategy selection yields meaningful gains over any fixed approach.

For **InstructBLIP**, Octopus generalizes effectively: MMHalBench score improves from 1.84 to **2.31** and HalRate from 0.64 to **0.49**; AMBER CHAIR drops from 8.4 to **6.1** with Coverage rising from 46.4 to **48.5**. Individual fixed CD methods on InstructBLIP show inconsistent or degraded gains (VCD even decreases MMHalBench score from 1.84 to 1.75), further illustrating the pitfall of applying a fixed strategy uniformly.

Notably, Octopus achieves competitive or superior performance compared to fine-tuning methods (HALVA, HACL, POVID, HA-DPO, EOS) *without modifying any LVLM parameters*, demonstrating its strong practical value.

Table 1. Comparison with the state-of-the-art methods for the generative task across three datasets. For reference, we also provide the results of fine-tuning LVLMs. † signifies results reproduced with the official implementation codes.

LVLMs	Method	MMHalBench		Object HalBench		AMBER			
		Score ↑	HalRate ↓	$CHAIR_s$ ↓	$CHAIR_i$ ↓	CHAIR ↓	Cover. ↑	HalRate ↓	Cog. ↓
Referenced Results (Not Directly Comparable)									
GPT-4V [1]	Base	3.49	0.28	13.6	7.3	4.6	67.1	30.7	2.6
LLaVA-v1.5 -7B[15]	HACL [10]	2.13	0.50	-	-	-	-	-	-
	POVID [39]	2.08	0.56	48.1	24.4	-	-	-	-
	EOS [37]	2.03	0.59	40.3	17.8	5.1	49.1	22.7	2.0
	HA-DPO [38]	1.97	0.60	39.9	19.9	6.7	49.8	30.9	3.3
	HALVA [23]	2.25	0.54	-	-	6.6	53.0	32.2	3.4
Comparable Results									
LLaVA-v1.5 -7B [15]	Base	1.59	0.72	25.0	9.2	8.0	44.5	31.0	2.2
	+ LCD [17]	-	-	61.0	16.1	-	-	-	-
	+ ICD [30]	-	-	47.4	13.9	-	-	-	-
	+ OPERA [9]	2.15	0.54	45.1	22.3	-	-	-	-
	+ VCD [11]	1.96†	0.64†	23.6†	8.4†	6.7	46.5	27.8	2.0
	+ M3ID [7]	2.14†	0.61†	23.2†	7.3†	6.0	48.9	26.0	1.5
	+ AVISC [33]	2.19†	0.59†	22.1†	7.8†	6.3	46.6	25.6	2.0
	+ Octopus(Ours)	**2.61**	**0.50**	**20.8**	**6.6**	**4.8**	**49.2**	**23.4**	**1.2**
Instruct-BLIP [6]	Base	1.84	0.64	0.7	9.1	8.4	46.4	31.1	2.6
	+ LCD [17]	-	-	17.4	10.7	-	-	-	-
	+ ICD [30]	-	-	15.2	8.0	-	-	-	-
	+ OPERA [9]	-	-	16.6	6.8	-	-	-	-
	+ VCD [11]	1.75†	0.64†	0.8†	8.9†	7.6	47.7	29.9	2.2
	+ M3ID [7]	1.70†	0.65†	0.9†	7.6†	6.9	47.2	27.5	2.2
	+ AVISC [33]	2.03†	0.59†	0.7†	8.3†	6.7	46.7	28.0	2.6
	Octopus (Ours)	**2.31**	**0.49**	**0.5**	**6.8**	**6.1**	**48.5**	**22.2**	**1.3**

6.3. Main Results: Discriminative Tasks

Table 2 presents results on discriminative benchmarks for LLaVA-1.5-7B and InstructBLIP. On AMBER Discrimination, Octopus substantially outperforms all baselines: Accuracy improves from 67.00 to **76.70** (+9.7 points) and F1 from 71.10 to **82.70** (+11.6 points), surpassing the best individual CD method AVISC (70.70/75.45) by a wide margin. On POPE-MSCOCO (ALL), Octopus achieves 85.79% accuracy and 83.44 F1, surpassing the baseline (82.04/80.42) and the best individual method AVISC (83.39/81.01). Gains are consistent across all sampling strategies: Random (87.51/85.40), Popular (85.20/84.19), and Adversarial (82.22/81.44).

For InstructBLIP, Octopus achieves 74.00/79.70 on AMBER Discrimination and 84.79/83.43 on POPE-MSCOCO (ALL), again establishing the best results among all compared methods and confirming strong cross-architecture generalizability.

6.4. Ablation Study

Table 3 presents an ablation study evaluating different subsets of CD strategies in conjunction with the Octopus decision module on the AMBER generative benchmark (LLaVA-1.5-7B). Row 1 is the base model (CHAIR 8.0, Coverage 44.5); Row 2 naively applies all three CD strategies without the Octopus decision module, achieving only modest improvement (CHAIR 6.9, Coverage 46.2). Rows 3–5 apply Octopus with different pairs of strategies: Str1+Str2 (CHAIR 5.5), Str1+Str3 (CHAIR 5.7), and Str2+Str3 (CHAIR 5.5)—each substantially better than the baseline, but inferior to the full system. Row 6, using Octopus with all three strategies plus the null action, achieves the best performance: CHAIR **4.8**, Coverage **49.2**, HalRate **23.4**, Cognition **1.2**. This confirms two key points: (1) the learned dynamic selection of Octopus consistently outperforms naive strategy combination (Rows 3–5 vs. Row 2), and (2) all three CD strategies contribute complementary value—removing any one degrades performance.

6.5. Design Choice Analysis

Table 4 compares alternative design choices for Octopus on the AMBER generative benchmark:

Reward Signal. Rows 1–2 compare different reward signals for constructing preference pairs: using Coverage score alone (Row 1) or averaging all metrics (Row 2). Our final

Table 2. Comparison with the state-of-the-art methods for the discriminative tasks across two datasets. † signifies results reproduced with the official implementation codes.

	AMBER		POPE_MSCOCO							
	Discrimination		Random		Popular		Adversarial		ALL	
	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1
LLaVA-1.5-7B[15]	67.00	71.10	83.77	81.94	82.57	80.86	79.77	78.47	82.04	80.42
+ICD [30]	-	-	**87.51**	83.28	83.15	83.91	79.13	80.41	83.26	82.53
+ConVis [20]	-	-	84.70	-	83.20	-	81.10	-	83.00	-
+OPERA [9]	-	-	84.40	-	83.40	-	81.20	-	83.00	-
+VCD [11]	67.30	71.10	85.43	83.99	83.17	81.94	80.27	79.49	82.96	81.81
+M3ID [7] †	67.25	70.90	86.13	81.85	82.07	80.77	79.50	78.15	82.57	80.26
+AVISC [33]	70.70	75.45	84.67	82.21	83.67	81.27	81.83	79.55	83.39	81.01
+Octopus (Ours)	**76.70**	**82.70**	**87.51**	**85.40**	**85.20**	**84.19**	**82.22**	**81.44**	**85.79**	**83.44**
InstructBLIP [6]	68.20	74.60	81.53	81.19	78.47	78.75	77.43	78.00	79.14	79.31
+ICD [30]	-	-	84.36	83.82	77.88	78.70	75.17	77.23	79.14	79.92
+OPERA [9]	-	-	84.57	83.74	78.24	79.15	74.59	76.33	79.13	79.74
+VCD [11]	69.65	75.90	82.03	81.56	79.13	79.20	77.23	77.72	79.46	79.49
+M3ID [7] †	69.05	75.25	82.33	81.53	80.90	80.42	78.53	78.49	80.59	80.15
+AVISC [33]	72.60	78.60	86.03	84.41	84.27	82.77	81.83	80.67	84.04	82.62
+Octopus (Ours)	**74.00**	**79.70**	**86.63**	**85.30**	**84.90**	**83.55**	**82.83**	**81.43**	**84.79**	**83.43**

Table 3. **Ablation study.** We select different numbers of contrastive decoding methods as candidates to demonstrate the effectiveness of our approach. "Str1, Str2, Str3" indicate CD strategy VCD [11], M3ID [7] and AVISC [33] in Sec. 4.2, respectively.

	Str1	Str2	Str3	Octopus	CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1					8.0	44.5	31.0	2.2
2	✓	✓	✓		6.9	46.2	26.1	2.2
3	✓	✓		✓	5.5	48.7	25.8	1.5
4	✓		✓	✓	5.7	48.2	25.3	1.5
5		✓	✓	✓	5.5	48.4	26.2	1.6
6	✓	✓	✓	✓	**4.8**	**49.2**	**23.4**	**1.2**

Table 4. The effects of different settings for our method.

		CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1	Cover	5.4	**50.1**	26.1	1.2
2	Average	5.0	49.8	23.0	1.4
3	Monte Carlo [31]	5.1	48.1	23.9	1.3
4	PPO [24]	5.8	47.5	**23.0**	1.5
5	Pooling [3]	6.2	44.8	25.0	1.6
6	Cross-atteition [26]	5.1	49.0	24.2	1.3
7	Ours	**4.8**	49.2	23.4	**1.2**

method (Row 7) uses CHAIR-based ranking and achieves better CHAIR (4.8 vs. 5.4/5.0) with competitive Coverage.

Training Algorithm. Rows 3–4 compare Monte Carlo [31] sampling and PPO [24] against DPO. DPO (Row 7) achieves superior CHAIR (4.8 vs. 5.1/5.8), validating our choice of DPO for its stability and effectiveness.

Aggregation Method. Rows 5–6 compare average pooling [3] and cross-attention [26] for aggregating hidden states into the decision token. Cross-attention (Row 6) substantially outperforms pooling (CHAIR 5.1 vs. 6.2), confirming that dynamic attention over the full hidden state sequence is important for capturing the context needed to select the appropriate CD strategy.

6.6. Qualitative Analysis

Fig. 5 provides a qualitative token-level analysis showing that Octopus assigns different CD actions to different generation steps within the same sequence, confirming that it has learned to identify and address the specific hallucination cause at each step. Fig. 6 compares generated descriptions across methods: individual fixed CD strategies may reduce some hallucinated objects while introducing new errors or degrading coverage, whereas Octopus consistently produces accurate descriptions that are both hallucination-free and coverage-preserving.

7. Conclusion

We presented **Octopus**, a dynamic contrastive decoding framework that alleviates hallucinations in Large Vision-Language Models by learning to select the most appropriate CD strategy at every generation step. By demonstrating that hallucination causes are fundamentally hybrid—varying across both samples and tokens—we motivated the need for per-token adaptive strategy selection. Octopus addresses this through a lightweight transformer block with a learnable decision token trained via DPO, while keeping the LVLM

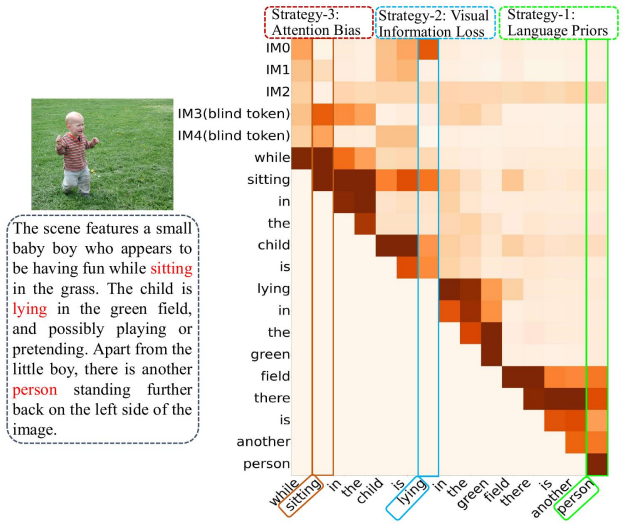

Figure 5. Token-level hallucination qualitative analysis. Different tokens within a generation are addressed by different CD strategies selected by Octopus.

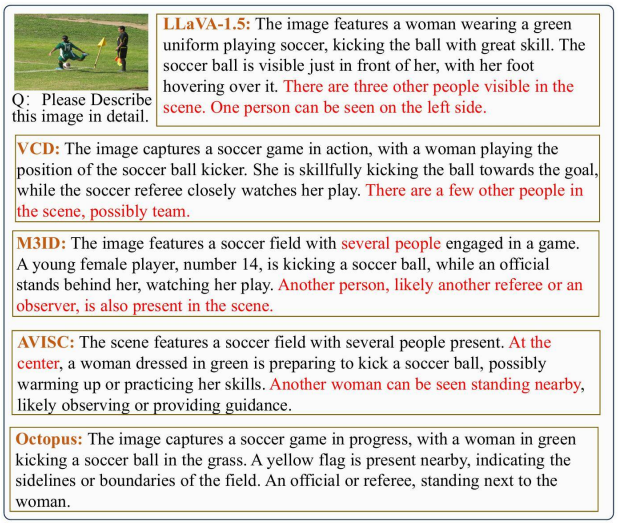

Figure 6. Comparison of generated descriptions with different CD strategies and Octopus. Fixed CD methods may improve some aspects while degrading others; Octopus consistently produces accurate, hallucination-free descriptions.

entirely frozen. Extensive experiments on LLaVA-1.5-7B and InstructBLIP across five benchmarks show that Octopus achieves state-of-the-art hallucination reduction on both generative and discriminative tasks, outperforming all fixed CD strategies and remaining competitive with fine-tuning approaches. Its plug-and-play design and extensible action space make it a practical and forward-looking solution for LVLM hallucination mitigation.

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